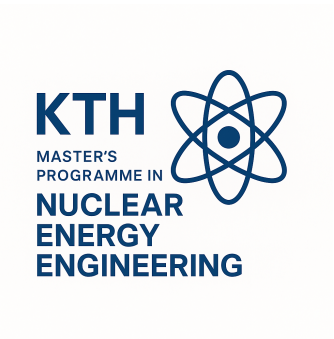




Stylistic Principles in Academic Writing

Nuclear Energy Engineering Master's Programme

Jan Dufek

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KTH Royal Institute of Technology
SCI | Physics | Nuclear Science & Engineering

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1 Introduction

This document summarises the stylistic principles of academic writing and includes a list of acronyms associated with respective stylistic rules. Review carefully all the rules and adopt them in your reports. The examiner will use these acronyms to mark the stylistic errors in your report. If you see an acronym in your reviewed report, it means you violated a principle. Carefully read the description of the principle to identify where your violated it.

Stylistic principles are not the only principles of academic writing. Other principles apply, and you must learn them before you start writing. You can learn them from our booklet "Guidelines for Writing a Scientific Report" by Dmitry Grishchenko and Ignas Mickus.

Once you assimilate the content of Grishchenko's booklet and this report, take a close look at any technical textbook written in English (for instance, Lamarsh's textbook that we use in the Reactor Physics course). Pay attention to how equations, plots, and tables are formatted, referenced, and how equations are integrated into sentences. Learn how to cite the statements and results of other authors.

If you wish to learn more about the principles and style of academic writing, you can do so using many textbooks that are freely available to you, e.g.: From Research to Manuscript: A Guide to Scientific Writing, A scientific approach to writing for engineers and scientists.

The most important stylistic principles are:

- **A report may contain sentences, floating elements, and titles, but nothing else.**
- **Floating elements are components that do not have a fixed position in the text and can move to optimise layout. Floating elements include tables, figures, diagrams and algorithms.**
- **Equations are not floating elements. Each equation must be fully integrated into a sentence. An equation cannot stand on its own.**
- **Each floating element must be numbered. Tables must have numbered titles, other floating elements must have numbered captions.**
- **Each floating element must be referenced from at least one sentence in the report.**

2 Report structure

[STR-CONTENT] The report must include title, date, the name of author(s), and sections Abstract, Introduction, Methods, Results, Discussion and conclusions. The section Methods may be split into multiple sections named differently.

[STR-ABSTRACT] The report must include an abstract. The abstract is a brief summary of the entire report. It should provide a brief description of the objective, methods, results, and conclusions. The abstract may contain multiple paragraphs, but it should not exceed half a page. The rest of the report should be written as if the abstract did not exist, meaning the abstract and the rest of the article must be completely independent and stand on their own.

[STR-INTRO] The introduction should state the purpose or task of the report and describe the structure of the report. The introduction should not contain any comments of the results.

[STR-METHODS] The report must include a section that provides sufficient theoretical background and a description of the methods, calculation routines, and numerical models used in your project. Your descriptions must be thorough enough to allow other researchers to reproduce your results. You may title this section "Methods" or something similar.

[STR-RESULTS] The report must include a "Results" section that presents the results in graphical, tabular, or textual form, depending on their nature.

[STR-DISCUSS] The report must include a "Discussion and conclusions" section, in which you should discuss your results and state your conclusions. Comment on whether you think the results are reasonable. If the results are not as expected, explain why you think so and what could be the reason for that.

[STR-CONSI] The style of titles of sections and subsections must be consistent. All words should either be capitalised or only the first word should be capitalised. Do not mix these styles in different titles.

3 Grammar

With AI grammar tools readily available, students should no longer submit reports written in broken English. Take advantage of these resources to ensure your grammar is correct before submitting your work.

[GR-ELEM] Only chemical element symbols are capitalised (e.g. U, Pu, H), not chemical element names (e.g. uranium, boron, hydrogen, nitrogen). Element names should be treated like common nouns.

[GR-MACHINE] Do not use the machine format for chemical or mathematical formulas. Instead of H2O write H_2O. Instead of U235 write ^{235}U.

[GR-ARTICLE] Make sure the definite and indefinite articles are used correctly. The indefinite article "a/an" is used for singular, countable nouns that have not been mentioned in previous text. The definite article "the" is used for countable nouns that have been mentioned before. Uncountable nouns do not use articles, unless exceptions apply. You may also omit all articles from titles if you do so consistently throughout the report.

The choice between “a” and “an” is based on pronunciation rather than spelling. For instance, “honour” begins with an “h” which is a consonant, but it is pronounced with a silent “h,” so we use “an” (e.g., “an honour”). The choice between using “a” and “an” thus depends on the initial sound of the word.

[GR-SPACE] A space is required after a full stop, comma, colon or semicolon. Do not add a space before these punctuation marks. Do not add extra spaces inside parentheses.

[GR-SHORT] Although short forms such as *“isn’t and can’t”* are grammatically correct, they are not used in technical reports.

[GR-PASSIVE] Sentences should be written in the active voice, not the passive voice. The passive voice is acceptable when the subject/actor is irrelevant. Most importantly, if you personally are the actor, definitely do not use the passive voice. Examples:

Correct: *We do not understand what went wrong. We obtained the data.*

Incorrect: *It is not understood what went wrong. Data were obtained.*

[GR-CONSI] Names must be typed consistently throughout the report. For example, the name of the same code should not be written as 'Serpent' in one sentence and

'SERPENT' in another.

[GR-AM/BRIT] Do not mix American and British English. Set your spellchecker to either American or British English and use it consistently.

[GR-SENTENCE?] Any text, including equations and lists, that is neither part of a title nor a floating element must belong to a sentence. Every sentence must have a subject and a verb, and it must end with a full stop.

[GR-FRENCH] When French-speaking students write in English, they might occasionally choose words that look similar to French ones—these are called *false friends* or *faux amis*. They exist in both languages but carry different meanings, which can cause confusion in academic writing. For example, they might use "material" instead of "nuclide", or "sensibility" instead of "sensitivity." Only use words that have the correct English meaning. Reading technical texts or textbooks written in native English helps reinforce correct usage. And using tools like domain-specific glossaries can catch those false friends.

[GR-SIMPLE] Keep sentences simple. Complex, over-complicated sentences are difficult to read and understand. Simplify sentences as much as possible. Complex, unreadable sentences are a sign of lazy writing. It takes effort to write well using simple sentences and words.

4 Paragraphs and sections

The text is structured into paragraphs that are either separated by vertical space or with the first word indented. Each paragraph begins with a statement sentence, and each subsequent sentence should develop, justify or expand upon the previous one. If a sentence does not relate closely to the previous sentence in the same paragraph, start a new paragraph with it. Similarly, each subsequent paragraph should relate to the previous one in the same section. If a paragraph does not relate closely to the previous one, start a new section.

[PAR-SHORT] A paragraph must contain more than one sentence. Either join the sentence to another paragraph, or expand it so that the paragraph contains at least three sentences.

[PAR-LONG] Paragraphs should not be too long. Break up long paragraphs. Each paragraph should deliver a single message. Once you have made your point, start a new paragraph. A paragraph spanning more than half a page is far too long.

[PAR-CONNECT] A paragraph should not be a collection of unrelated sentences. Each sentence must be connected to the previous one in some way. A paragraph begins with a statement that connects to previous paragraphs.

[SEC-CONNECT] A section should not be a collection of unrelated paragraphs. Every paragraph in a section must be connected to the previous one in some way way.

5 Equations, formulas, variables, numbers

Examples

The total energy E is proportional to mass m as

$$E = m \times \mathrm{c}^2, \tag{5.1}$$

where c is the speed of light in vacuum. Note that c in Eq. 5.1 is a constant and not a variable, and therefore it is typed in the standard text font.

Since a mole of identical atoms (with a mass of M grams) contains N_A atoms, the mass m of an individual atom is

$$m = \frac{M}{N_A}. \tag{5.2}$$

The momentum p of a particle is given by

$$p = mv,$$

where m is the particle's mass and v is its velocity. In classical physics, a particle's kinetic energy T is given by

$$T = \frac{mv^2}{2} = \frac{p^2}{2m},$$

which yields

$$p = \sqrt{2mT}. \tag{5.3}$$

[EQ-INTEGRATE] Equations are always part of a sentence. They cannot stand independently outside of sentences. An equation does not require a reference number unless it is being referred to from another sentence.

[EQ-WHERE] If the word 'where' is first word written directly below an equation to specify the variables in that equation, as in the above example, it must be written in lowercase (since it continues the sentence) and not indented (since only the first line of a paragraph is indented).

[EQ-VAR] Variables are always written in italics, regardless of whether they appear in an equation or in the text. The font style of a variable must be consistent wherever it appears: in the text, in an equation or in a plot.

[EQ-NOTVAR] Any term that is not a variable should not be written in italics. Constants, units and chemical elements, for instance, are not variables and should be written in regular font. Note that the "eff" in the neutron multiplication factor k_{eff} is not a variable either (it is a label), and should be typed in the regular font.

[EQ-MULTI] Do not use the asterisk symbol (*) for multiplication in equations and formulas. Use "$\times$" instead. Also, only use the dot operator ($\cdot$) for the dot product of two vectors. Using it elsewhere will confuse readers. When multiplying a scalar by a variable, use no operator or, preferably, use "$\times$" (the "\times" command in the LaTeX math mode).

[EQ-NUM] Numbers must be written in the correct regular or scientific format. The decimal point should be typed as a point, not a comma. Scientific notation is not written in machine format and does not include the characters "*, e, E or $\cdot$"; instead, it uses the character "$\times$". See Table 7.1 for examples of correctly and incorrectly written numbers.

[EQ-DECIMAL] The number of decimal places stated in numerical results should be reasonably small in relation to the associated errors or statistical deviations. There is no good reason to state a result with five or more decimal places when the statistical deviation or other uncertainties limit the accuracy of the result to one or two decimal places at most.

[EQ-WORDS] Do not use words (or even sentences) instead of variable names in equations. Instead, assign a short name — preferably a single-character name — to each variable, and use these inside the formulas and equations.

[EQ-DEFINE] All variables in an equation must be defined either in the text preceding the equation or immediately after it, preferably within the same sentence.

[EQ-MACHINE] Equations must be typed in the correct mathematical format. Do not copy and paste equations in machine format from your computer script.

[EQ-UNIQUE] A single variable must be given a unique name and style throughout the report (including plots). For example, if you write k_{eff} in one place, you cannot write the same variable as k, or K, or K_{eff} elsewhere. Check that the names and styles of all your variables are typed correctly in every plot.

6 Figures

Figure 6.1 serves as an example here. Note that the text on the axes and in the legend uses a font size similar to that used for normal text. The figure caption is always present and located beneath the figure, and the figure number is specified in front of the caption. Figures may or may not have titles. If the title is present then it is positioned above the figure, but the figure number is still in front of the caption. As the figure was generated in vector format, the curves and text remain sharp when zoomed in.

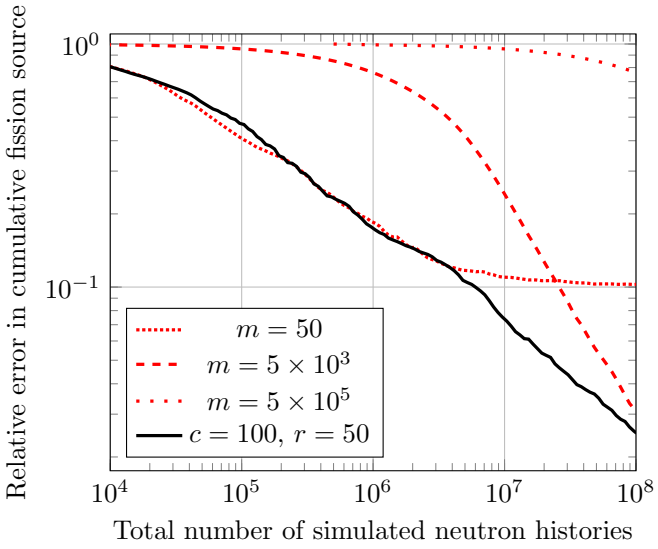


Figure 6.1: Convergence of the cumulative fission source in simulations with fixed and variable neutron population size.

[FIG-NUM] Every figure must be numbered.

[FIG-CAP] Each figure must have a caption. Captions should always be placed below the figure and end with a full stop.

[FIG-REF] Each figure must be referenced in a sentence. Figures cannot be included in the report unless they are referenced in a sentence.

[FIG-SMALL] Any text on the figures, including the axis labels, should be a comparable font size to normal paragraph text. The figures must be legible when printed or viewed as a PDF without the need to zoom in.

[FIG-LABEL] The axes of the figure must be labelled.

[FIG-UNITS] The units must be present on the figure axis/axes unless the variable has no unit.

[FIG-BITMAP] When exporting a plot from your calculations, choose a vector format rather than a bitmap format. Vector formats, including PDF, EPS and SVG, enable you to zoom into the plot without losing resolution. Bitmap formats, such as JPG, PNG or GIF, are exported at a fixed resolution and exhibit compression artefacts, so are not suitable for plots. The only exception to this rule is when graphics, such as photos, are only available in bitmap format.

[FIG-COPYRIGHT] All graphical elements (photos, drawings, plots, etc.) in your thesis must be copyrighted. If you cannot acquire the copyright, produce your own drawing instead.

7 Tables

Table 7.1 serves as an example here. Tables do not use captions beneath the tables; instead, they use titles above the tables. The table number is specified in front of the title. Tables offer a great way to present parameters and results of your calculations and measurements in an organised way.

Table 7.1: Examples of correctly and incorrectly written numbers

	Regular format	Scientific format
Correct	0.3; 1.; .2	2.3×10^{5}; 1.4×10^{-23}
Incorrect	2,3	2.3e5; 2.3E-5; $2.3 \cdot 10^{5}$; $2.3*10^{5}$

[TAB-TITLE] Each table must be numbered and titled, with the title placed above the table. See Table 7.1 for an example. As it is a title, it does not end with a full stop.

[TAB-UNITS] All numerical data in the table must have specified units, either in the column/row headers or explicitly in the title.

[TAB-REF] Every table must be referenced at least once in the text.

8 References

Tables, figures, equations and sections must be numbered so that they could be referenced from text by the number. You can write, for instance, "*the data are in Table* 7.1". Note that "Table 7.1" is the name of the table, and so it is capitalised. If you address some tables in general then the word "tables" is not capitalised. Note that there is a space between the word "Table" (or "Tab.") and the number. The same rules apply for figures, equations, sections and any other numbered item, e.g. "*Assignment 1, Point 1*". Reference numbers may or may not be enclosed in parentheses.

Example

Results are summarised in Table 1. Results are given in Tab. 1. We conclude that Eq. (1) is incorrect. We assume that Eqs. 1-5 are correct. We assume that Equations (1-5) are correct. Our results are given in Fig. 1. Our results are stated in Figure 2. We generated several figures in Sec. 3.

[REF-CAP] The reference name must be capitalised, written e.g. as "Eq. (2)", "Fig. 1", "Sec. 3", "Tab. 2".

[REF-SPACE] The reference name must include a blank space between the name and the number, so it cannot be written as e.g. "Fig.3".

[REF-PERIOD] When using a short reference name, such as "Fig. 1" instead of the full name "Figure 1", the short name must include a full stop and cannot be written as "Fig 1".

[REF-NUMBER] The referenced item must be numbered. You cannot write "*results are given in the Table below*" or *"results are given in the next table*".

[REF-ARTICLE] Articles are not used when referencing by name. For example, you cannot write "*results are in the Table 1*".

[REF-MISSING] It must be clear which table, figure or equation you are referring to.

9 Lists

A list consists of one or more sentences that have been visually broken into multiple lines, but which still have all the attributes of proper sentences. If you compacted the list, it would read as normal text. The text must still obey grammar rules.

A list example that contains multiple sentences:

We make the following assumptions:

- **Neutron energy-dependent cross sections are approximated by tables (obtained via the Janis database).**
- **The geometry treatment uses the delta tracking method.**

A list example that contains a single sentence:

We assume that:

1. **neutron energy-dependent cross sections are approximated by tables (obtained via the Janis database),**
2. **the geometry treatment uses the delta tracking method.**

Another list example that contains a single sentence:

The following colours were visible:

- **blue,**
- **green,**
- **yellow.**

[LIST-WRONG] A single-sentence list should begin with a capitalised introductory phrase, followed by non-capitalised numbered or bulleted statements separated by commas, and end with a full stop. A multi-sentence list must contain capitalised, numbered (or bulleted) sentences that end with full stops.

10 Citations

Citations allow you to give credit to the original authors of theories, methods or data. This enables readers to verify your claims and conduct further research. Every non-obvious statement you make must either be justified or referenced properly (i.e. to another report, article, book, or person). There are various citation formats. Citations can be numbered and referenced via the number or by the author's name or acronym. Bibliography or References should be added to the end of the report as the last unnumbered section. A standard bibliography item format includes the following: author(s); title; journal or publisher; volume; page numbers; year of publication. See the bibliography on the last page of this report.

Examples

The Monte Carlo method can be used to solve neutron transport problems [1].

See Fig. 1 in [2] for the convergence behaviour of the method.

[CITE-MISSING] Every non-obvious statement you make must either be justified by the preceding text or be properly referenced (i.e. referenced to another report, article, book or person).

[CITE-END] If a statement is entirely based on a reference, place the citation at the end of the sentence before the full stop.

[CITE-ALL] All cited items must appear in the bibliography, and that all items in the bibliography must be referenced in the report.

Bibliography

[1] J. P. Both, A. Mazzolo, Y. Pénéliau, O. Petit, and B. Roesslinger. *User Manual for Veriosn 4.3 of the TRIPOLI-4 Monte Carlo Particle Transport Computer Code*. Commissariat à l'Énergie Atomique (CEA), Saclay 91191, 2003.

[2] R. J. Brissenden and A. R. Garlick. Biases in the estimation of k_{eff} and its error by Monte Carlo methods. *Ann. Nucl. Energy*, 13(2):63, 1986.